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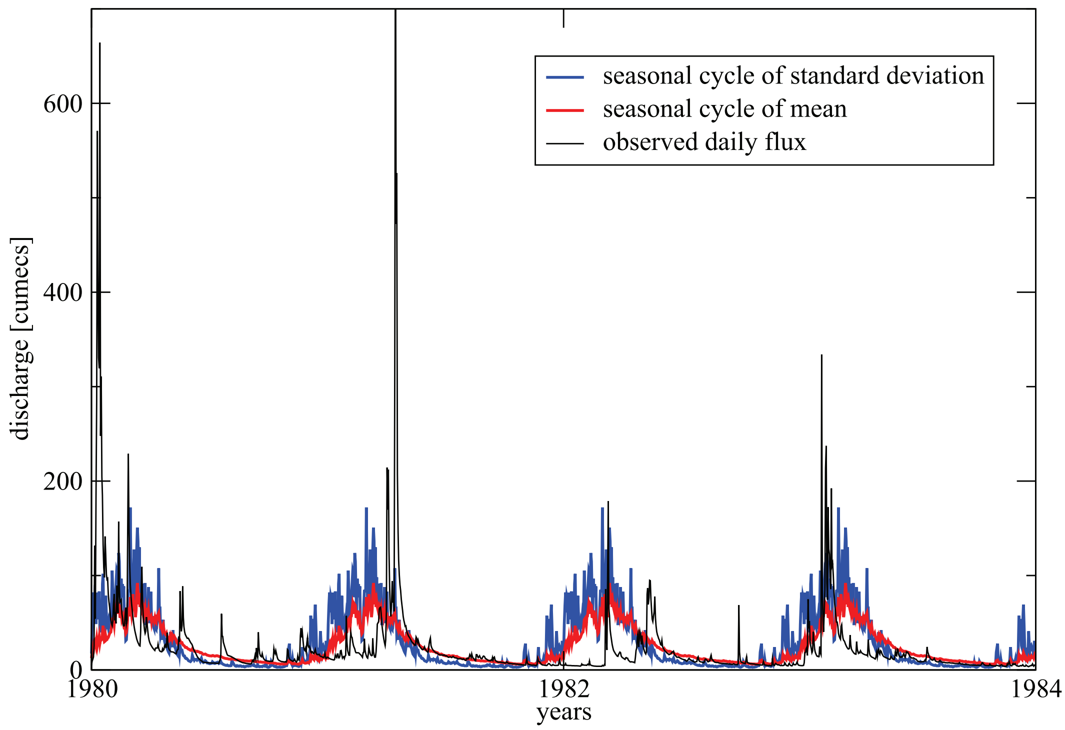
IN EXTREMIS
Extremes, Trends, and Correlations in Hydrology and Climate

SPIN; Version: April 15, 2010

Springer-Verlag
Berlin Heidelberg New York
Barcelona Hong Kong London
Milan Paris Singapore Tokyo


seasonal cycle of standard deviation
seasonal cycle of mean
observed daily flux
discharge [cumecs]
600
400
200
0
1980
1982
1984
years

13. Seasonality effects on nonlinear properties of hydrometeorological records

Valerie N. Livina, Yosef Ashkenazy, Armin Bunde, and Shlomo Havlin

Climatic time series in general, and hydrological time series in particular, exhibit pronounced annual periodicity. This periodicity and its corresponding harmonics affect the nonlinear properties of the relevant time series (i.e., the long-term volatility correlations and the width of the multifractal spectrum) and thus have to be filtered out before studying fractal and volatility properties. We compare several filtering techniques and find that in order to eliminate the periodicity effects on the nonlinear properties of the hydrological time series, it is necessary to filter out the seasonal standard deviation in addition to the filtering of the seasonal mean, with conservation of linear two-point correlations . We name the proposed filtering technique “phase substitution”, because it employs the Fourier phases of the series. The obtained results still indicate nonlinearity of the river data, its strength being weaker than under previously used techniques.

13.1 Introduction

To better understand the climate dynamics, the analysis of the observed climate time series is essential. Their statistical properties provide, e.g., with the average state of the climate system, the order of the natural fluctuations around this state, and the climate trends [13.23].

One basic way to quantify the statistical properties of a time series is through the probability distribution function (pdf) of the data. The pdf, however, is invariant under shuffling of the time series and does not reflect the temporal statistical properties (those related to the time ordering of the time series). For instance, under shuffling of the time series, the probability dis-

◄ **Fig. 13.0.** Johnstone river flux with seasonal cycles of mean and standard deviation tribution remains unchanged, while the autocorrelation function is affected. Still, estimation of the forms of the pdf may help, for example, to quantify the relative occurrence of extreme climate events [13.2].

Time ordering of a climatic record may teach us about the dynamical properties of the climate system [13.13]. Pronounced periodicities, like the annual periodicities of temperature and river flow, can easily be tracked using the Fourier frequency spectrum. The harmonics of the annual periodicities may provide useful information regarding the "morphology" of the annual cycle, and usually are indicators of the nonlinear underlying process (or of the nonlinearity of the time series). The periodicities are usually linked to the Fourier amplitudes; however, as is demonstrated in this paper, some nonlinear properties (such as the periodicity in the moving variance of the time series) may be associated with the Fourier phases [1].

In earlier works, the hydrological time series were studied either in 'raw' format (no filtering of periodicities), or after removing seasonal cycle of mean and/or standard deviation. Here, we present a comparison of the scaling properties of river records after applying various types of filtering; we also propose a new filtering technique and discuss its effects.

Fig. 13.0 shows a representative hydrological record (river flux of the Johnstone river) with pronounced annual periodicity in mean and standard deviation. For this river, we show in Fig. 13.1 that after subtracting from each daily value its corresponding multi-year daily average (deseasonalising), the obtained volatility time series still has periodicities. This residual effect is due to the fact that in some seasons (e.g., summer and autumn) the river flow fluctuations are small, while in the other seasons they are very pronounced. This remaining periodicity drastically influences the measurement of the nonlinear scaling properties of river flux time series; therefore, the goal of the present paper is to compare several techniques that might cope with such periodicities. We show that it is necessary to remove the periodicities which are associated with Fourier phases, as well as those associated with the Fourier amplitudes.

The complex discrete Fourier transform of a series $x_\ell,\ \ell = 1, \ldots, N$,

$$X_\omega = \sum_{\ell=1}^{N} x_\ell e^{-i(2\pi\omega/N)\ell}, \quad \omega = 1, 2, \ldots, N, \tag{13.1}$$

allows the study amplitudes $|X_\omega|$ and phases ϕ_ω of the series:

$$X_\omega = |X_\omega| e^{i\phi_\omega}. \tag{13.2}$$

Fourier amplitudes and phases may be associated with linear and nonlinear properties of the series respectively. By *linear* we refer to a process that appears

[1] We use term "linear/nonlinear time series" as a synonym for the linear/nonlinear underlying process that generated the time series.

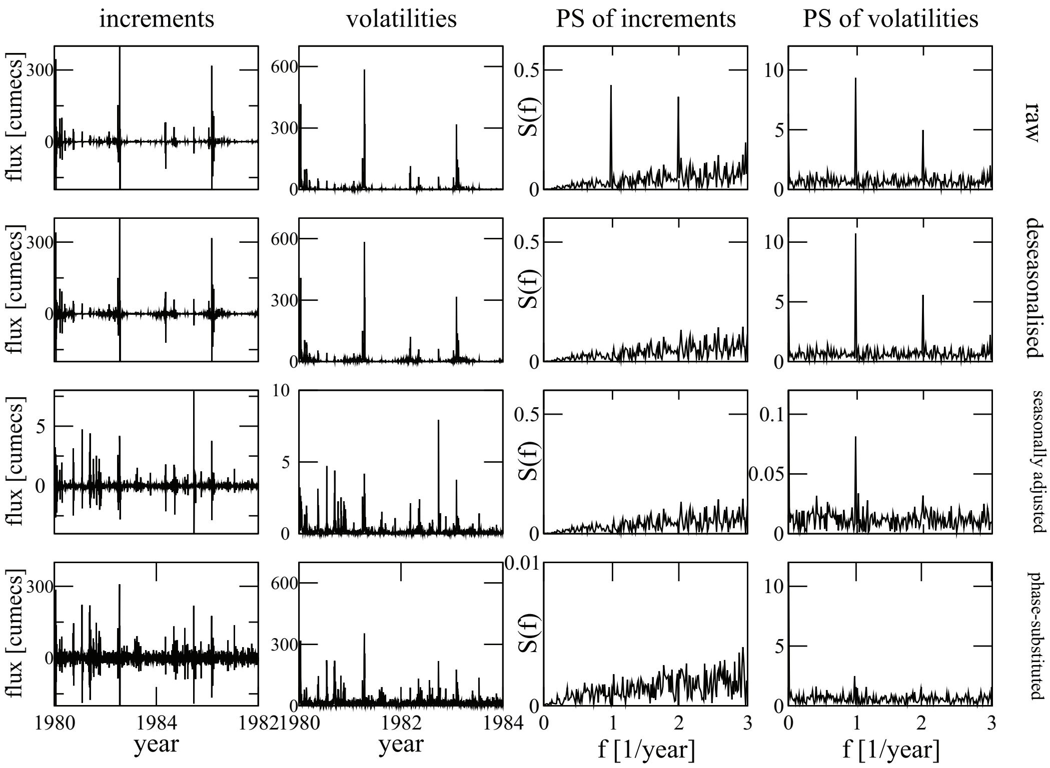


Fig. 13.1. Increments, volatilities, power spectra of increments, and power spectra of volatilities for Johnstone river: raw, deseasonalised, seasonally adjusted, and phase-substituted data (see detailed explanation of the filtering techniques in Methodology). The conventional deseasonalising procedure eliminates the periodicity in the power spectrum of increments—much the same as the phase-substitution does—but fails to eliminate the periodicity in the power spectrum of volatilities of increments. The new phase-substitution filtering method does eliminate the periodicity peak in the power spectrum of volatilities: compare the panels in the last column.

only in the power spectrum and in the probability distribution but is independent of the Fourier phases; otherwise the process is defined as *nonlinear*. In other words, if the statistical properties, including those of large moments, remain unchanged after randomizing the Fourier phases, the process is linear, while in the case where the statistical properties of large moments are modified after randomizing the Fourier phases, the process is defined as nonlinear. By using Schreiber's test for nonlinearity [13.20] , one ensures that a linear time series that was transformed by a monotonic and static nonlinear function will still be regarded as linear; for more details see [13.20]. Accordingly, Schreiber's test verifies whether the null hypothesis for linearity is rejected or not. To ensure stationarity, we concentrate mainly on the increment series and not on the original series.

Temporal correlations of climate records have been studied extensively. In the '50s, Hurst [13.8] reported long-term correlations in river flow records, which was confirmed in further studies [13.7, 13.11]; long-term temporal correlations were also found in other climatological time series, such as temperature records [13.9,13.10]. The surprising finding is that even time-distant events are statistically correlated, and that these long-term correlations may be quantified by a single scaling exponent H in the corresponding fluctuation function.

This scaling exponent, sometimes called Hurst exponent, is related to the power spectrum scaling exponent β by $H = (1+\beta)/2$ and, following Schreiber's definition of linearity, quantifies the linearity of the time series. It was found that even the higher moments temporal correlation functions of hydrological time series also obey scaling laws [13.12,13.28], where these scaling laws reflect the nonlinearity of the processes generating the time series. In the case of a linear process, all different moments may be characterized by a single exponent, and the series is *monofractal*, while a nonlinear time series has a spectrum of exponents called the *multifractal spectrum*—the stronger the nonlinearity, the broader the multifractal spectrum.

To measure the scaling exponents of the fluctuations of a climate record, the periodicities should be filtered out first; otherwise, the verification and estimation of the scaling exponents may be affected by these periodicities. When dealing with two-point correlation functions (the 2nd moment), it is sufficient to filter out the periodicities from the power spectrum. However, when dealing with nonlinearities, or general moment scaling exponents, it is also necessary to filter out the periodicities associated with these moments. In Fig. 13.1, we show that even after eliminating the annual periodicity and its harmonics, the series still has periodicities which are not captured by the power spectrum but rather lie in the Fourier phases. Our point is that an inadequate filtering procedure does not eliminate these effects of periodicities, and thus may lead to the observation of an artificial nonlinearity in the data. We show that by removing these higher moment periodicities, the nonlinearity of the time series weakens and the scaling range is extended.

Here we compare five approaches in studying hydrological time series:

Approach 1: 'Raw' time series, no filtering applied.
Approach 2: The seasonal cycle of mean values is filtered out from the original time series.
Approach 3: The seasonal cycles of mean and standard deviation are filtered out from the original time series.
Approach 4: The new technique (described below) — 'phase substitution' — is applied to the original time series.
Approach 5: The new technique — 'phase substitution' — is applied to the *increments* of the original time series.

Taking increments $x_{\ell+1} - x_\ell$ in Approach 5 aims at the nonstationarities of the hydrological time series (scaling exponent $H > 1$) and is discussed in details in section 13.4.

The paper is organized as follows. First we review briefly the earlier filtering techniques and discuss the problems with some of them. We then propose a technique that filters out the periodicity from both the power spectrum and the volatility series (given a time series x_ℓ, the volatility series is defined as $|x_{\ell+1}-x_\ell|$). Next, we demonstrate the use of this technique on hydrological time series and show that the nonlinearity weakens (or the multifractal spectrum width becomes narrower) when using the proposed technique, as well as when using some of the other procedures that can filter out the periodicity in the standard deviation. We then summarize the results and draw some conclusions.

13.2 Methodology

13.2.1 Detrended Fluctuation Analysis and Multifractal Detrended Fluctuation Analysis

Long range correlated time series may be characterized by scaling laws with specific scaling exponents. It is possible to show that the power spectrum $S(\omega) \equiv |X_\omega|^2$ of long-range correlated time series scales with the frequency ω as $S(\omega) \sim \omega^{-\beta}$. However, trends that exist in climate records may cause inaccuracies in measuring this exponent, in particular, at low frequencies, and thus more advanced techniques that systematically exclude trends from the data have been developed. These techniques include wavelets [13.1, 13.9] and the Detrended Fluctuation Analysis (DFA) [13.6, 13.26].

In recent years, the DFA method has become a widely used tool for measuring scaling exponents of natural time series. It was applied, e.g., to DNA [13.21] and heart-rate sequences [13.3, 13.5, 13.27], financial time series [13.17, 13.18], and climate records [13.4, 13.9]. In this method, one considers fluctuations of the cumulative sum ("profile") $Y_m = \sum_{\ell=1}^{m} x_\ell$ of the original series x_ℓ, $\ell = 1, \ldots, N$. We divide the time axis into nonoverlapping windows of size s. Next, within each window, we calculate the best polynomial fit and determine the variance

$$F^2(\nu, s) = \frac{1}{s} \sum_{\ell=1}^{s} \left[Y((\nu-1)s+\ell) - y_\nu(\ell)\right]^2 , \tag{13.3}$$

where $y_\nu(\ell)$ is the least square polynomial fit of order k in segment ν of size s. Finally, we average $F^2(\nu, s)$ over all windows ν with fixed size s to obtain the fluctuation function $F(s)$.

When a series obeys scaling, the fluctuation function $F(s)$ follows a power-law:

$$F(s) \sim s^\alpha , \tag{13.4}$$

where α is the DFA fluctuation exponent (equivalent to the Hurst exponent H). We calculate $F(s)$ for time scales between $s = 5$ and $s = N/4$, where N is the length of the series. We are mainly interested in the long-term persistence, i.e., in scales beyond one year (denoted by vertical dashed lines in Fig. 13.2.

For uncorrelated records $\alpha = 0.5$, while for long-term correlated (persistent) records, where the autocorrelation function decays with time lags as $C(s) \sim s^{-\gamma},\ 0 < \gamma < 1$, the exponents are related as $\alpha = 1 - \gamma/2$. In DFAk, trends of order $k-1$ are eliminated (k^{th} order trends from the profile). The fluctuation exponent α is also related to the power spectrum scaling exponent $S(f) \sim 1/f^{\beta}$ by $\beta = 2\alpha - 1 = 1 - \gamma$.

To study the multifractal properties of the hydrological time series, we use the Multifractal DFA (MFDFA) method [13.24], which is a generalization of the DFA method. In the MFDFA procedure, similarly to the conventional DFA, we calculate the generalised fluctuation function for arbitrary moments q (for the conventional DFA, $q = 2$):

$$F_q(s) = \left\{ \frac{1}{N_s} \sum_{\nu=1}^{N_s} [F^2(\nu, s)]^{q/2} \right\}^{1/q}, \quad F_q(s) \sim s^{h(q)}, \tag{13.5}$$

where $h(q)$ is the generalized Hurst exponent. For monofractal time series, $h(q)$ does not depend on q and is constant. For positive values of q, $h(q)$ characterises the scaling behaviour of segments with large fluctuations; for negative values of q, $h(q)$ is associated with the scaling behaviour of the segments with small fluctuations. For our analysis, we use MFDFA with 3^{rd} order polynomials (MFDFA3), thus eliminating quadratic trends in the original data.

Exponents $h(q)$ can be related to the standard scaling exponent $\tau(q)$ [13.15] by

$$\tau(q) = qh(q) - 1. \tag{13.6}$$

One often characterizes the multifractality by the singularity spectrum $f(\alpha)$, which is related to $\tau(q)$ via the first-order Legendre transform [13.15]:

$$\alpha = \tau'(q), \quad f(\alpha) = q\alpha - \tau(q). \tag{13.7}$$

The width of the spectrum $f(\alpha)$ characterizes the strength of multifractal effects in the data. For monofractal data, the spectrum $f(\alpha)$ collapses into a single point, and both functions $\tau(q),\ h(q)$ are linear.

13.2.2 Volatility Correlations and Surrogate Test for Nonlinearity

We define the volatility series as $|\Delta x_\ell| = |x_{\ell+1} - x_\ell|,\ \ell = 1, 2, \ldots, N-1$ [13.19, 13.27]. Correlations in the volatilities reflect the way the magnitudes of fluctuations are arranged; when the fluctuations are clustered into patches of large and small fluctuations, the volatility series is correlated. When the fluctuations are more homogeneous, the volatility series is less correlated.

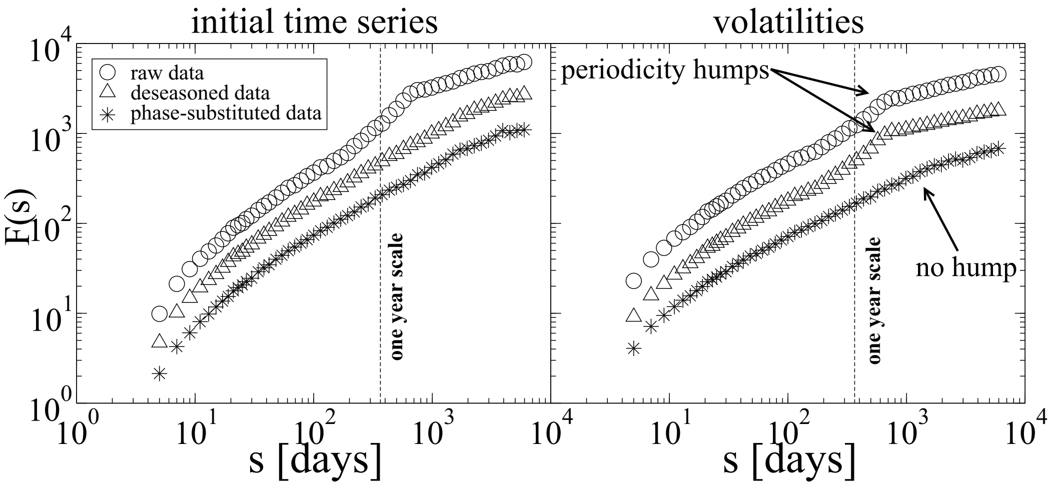


Fig. 13.2. DFA3 results for Johnstone river. Left panel: curves for raw (circles), deseasonalised (triangles), and phase-substituted (stars) data. Right panel: the same for volatilities. Note that conventional deseasonalising eliminates the periodicity hump in the initial data (like the phase-substitution procedure), but fails to eliminate the periodicity hump in the DFA curve of the volatility series, where the phase-substitution works well.

Since the information regarding the nonlinearity of the time series is stored in the Fourier phases, it is possible to generate linearized surrogate time series by randomizing the Fourier phases. However, when randomizing the Fourier phases, the histogram of the original time series may be affected. A method to preserve both the histogram and the power spectrum of the original time series was introduced by Schreiber and Schmitz [13.20].

In a surrogate data test, the NULL hypothesis is that the series under consideration is linear. The surrogate data has the same pdf and almost the same power spectrum as the original series, but with random Fourier phases. If a statistical measure obtained for the original series is significantly different from that of the surrogate data, the NULL hypothesis is rejected and the series is considered to be nonlinear. Applying the surrogate test, we compare periodic volatility and volatility correlations of initial data and surrogates to measure the nonlinearity.

We apply DFA3 to study correlations in the volatility series. In our previous work on river records [13.22], we showed that in the time range between 1 week and 1 year, the volatility fluctuation exponent $\alpha \sim 0.75$, and for time scales larger than one year $\alpha \sim 0.65$. When applying the surrogate data test for nonlinearity, the volatility exponent decreases to $\alpha \sim 0.55$ for time scales above one week. The results of surrogate data test, i.e., the decrease in the volatility exponent from large value to ~ 0.5, indicate the existence of nonlinearity in the original river flow time series (see figures in [13.22]).

Since the surrogate test should be applied to stationary time series (i.e., with DFA exponent smaller than one), we will apply the surrogate data test on the increment series, $\Delta x_\ell = x_{\ell+1} - x_\ell$ when considering hydrological time series that are nonstationary with $\alpha > 1$ for time scales smaller than one year.

13.3 The Way Periodicities Affect the Estimation of Nonlinearities; Conventional Seasonal Detrending

By the conventional deseasonalising (seasonal detrending) we refer to elimination of the general climatic periodicity, i.e., subtracting the average annual cycle: $x_\ell - \langle x \rangle_{day}$, where $\langle x \rangle_{day}$ indicates the average of the particular day (or month in the case of monthly data) over the years.

More advanced conventional method of seasonal detrending is "seasonal adjustment", in which not only seasonal cycle of mean values is subtracted, but also new series is divided by the seasonal cycle of standard deviation (see, e.g., [13.16]):

$$\frac{x_\ell - \langle x \rangle_{day}}{\sqrt{\langle x^2 \rangle_{day} - \langle x \rangle^2_{day}}}. \tag{13.8}$$

In Fig. 13.2, the DFA3 curves for raw (circles) and conventionally deseasonalised data (triangles) are compared, left panel for the initial time series and right panel for the volatility series. One can see that though the conventional deseasonalising copes well with the periodicity hump in the initial data (left panel), in the the curves for volatilities (right panel) the deseasonalising is not enough, and the periodicity crossover is still present and certainly influences the estimation of the scaling exponent.

It can be shown that the seasonal adjustment (13.8) alters the scaling exponent of the filtered data. Therefore, the fragile two-point correlations should be monitored in the course of processing the data. For more extended nonlinear analysis, such as the multifractal and volatility analyses, sindexAnalysis!volatility both conventional filtering techniques, deseasonalising and seasonal adjustment, are insufficient.

13.4 A New Method for Filtering Out Periodicities

In a recent study [13.11], it was shown that the river data are nonstationary at short time scales (fluctuation exponent above 1 in time scale up to several months) and are long-term correlated with $\alpha \sim 0.75$ in the asymptotic time scale. Therefore, to eliminate the nonstationarities, we consider the increment series $\Delta x_\ell = x_{\ell+1} - x_\ell$, $\ell = 1 \ldots N-1$. By this, the fluctuation exponent is reduced by 1. This linear transformation (discrete differentiation) is harmless in the sense of correlations: once the series has been integrated back, the original

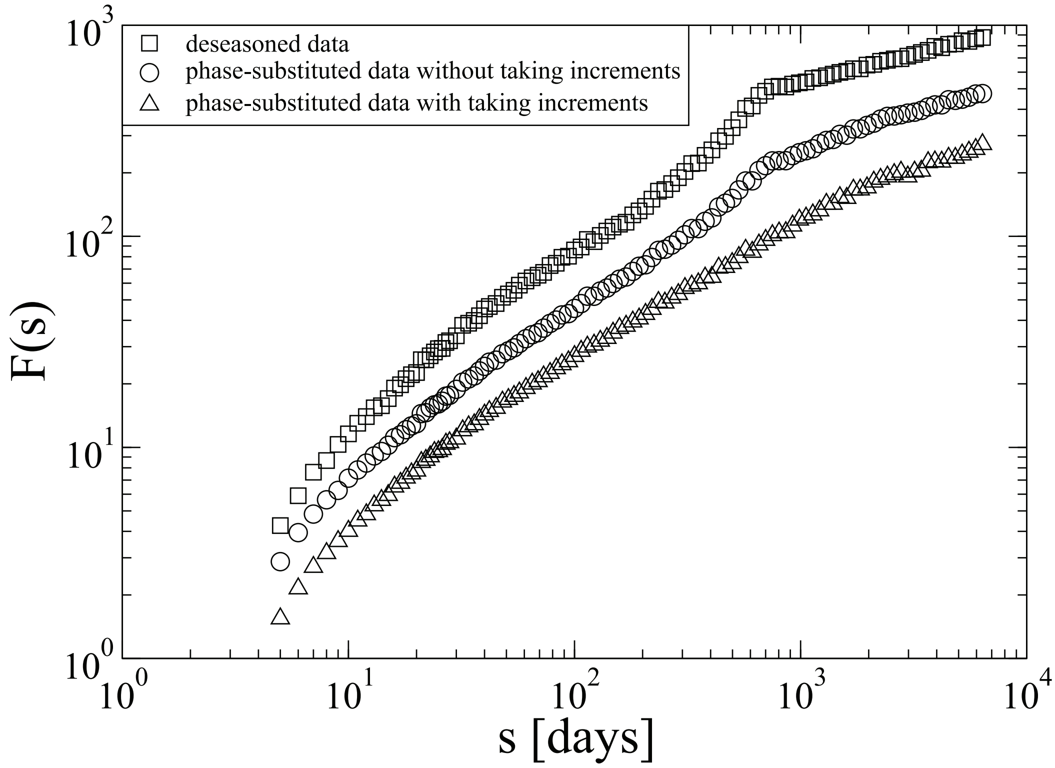


Fig. 13.3. Motivation for analysing the increments of the data: DFA3 results for volatilities of the Johnstone river. Applying phase-substitution on the initial data as compared to the increments with back integration shows that there is some periodicity which still cannot be eliminated due to the nonstationarities of the record.

fluctuation exponent is restored. The obtained increment series is stationary (fluctuation exponent is less than one), and one can use it for further analysis (for instance, apply conventional deseasonalising), and then integrate the series back. This would provide the series with removed periodicity from the second moment curve (two-point correlations) and improve the result of the linear correlation analysis.

If the conventional deseasonalising is to be applied, we can consider either the initial series or the integrated increments. The importance of incrementing the data becomes more pronounced in the case of the seasonal adjustment which is essential for multifractal and volatility analyses. In Fig. 13.3, where the volatility analysis is performed on deseasonalised and phase-substitution filtered data (the latter filtering described below), one can see that applying the new filtering onto data without taking increments still provides some retained periodicities, whereas the volatility curve for the phase-substitution filtering applied to the increment series with further integration provides a flatter region in the seasonal scale of the curve.

To account for the above considerations, we process the data as follows.

(I) Possible nonstationarities in the series are eliminated by taking the increments $\Delta x_\ell = x_{\ell+1} - x_\ell$, $\ell = 1 \ldots N$.
(II) We filter the time series by subtracting the daily (or monthly) climatological average (periodic seasonal average).
(III) To further eliminate the volatile periodicity of fluctuations, we divide the obtained series by the annual cycle of the climatological standard deviation, see Equation (13.8). Combining steps II and III is seasonal adjustment.
When applying seasonal adjustment, which is a nonlinear operation, we have observed that the correlations in the increment series may weaken (possible procedure artefacts). In order to preserve both two-point correlations and nonlinear correlations (that are stored in the Fourier phases), we perform one more step.
(IV) Since the linear properties are linked to the power spectrum and the nonlinear properties are linked to the Fourier phases, we now combine the Fourier amplitudes obtained from the series in step II with the Fourier phases obtained in step III. We denote the Fourier transform of the deseasonalised data from step II as $X_\omega^{(1)} = |X_\omega^{(1)}|e^{i\phi_\omega^{(1)}}$, and the Fourier transform of the seasonally adjusted data from step III as $X_\omega^{(2)} = |X_\omega^{(2)}|e^{i\phi_\omega^{(2)}}$. The new Fourier transform is formed as $W_\omega = |X_\omega^{(1)}|e^{i\phi_\omega^{(2)}}$ and transforms back (inverse Fourier transform):

$$z_\ell = 1/N \sum_{\omega=1}^{N} W_\omega e^{2\pi i \ell \omega / N}, \tag{13.9}$$

thus providing the final filtered series z_ℓ. After that, we integrate the data back to restore the fluctuation exponent which was reduced by 1 when considering the increment series in step I. In this way, both linear and nonlinear scaling properties are preserved, and the periodicities are eliminated.

The obtained "phase-substituted" data are further studied in the next section.

13.5 Results

13.5.1 Tests of Artificial Data

To perform an extended test of the technique on simulated time series with known properties, we have generated four ensembles of artificial data, using (i) monofractal long-term correlated data [LTC] and (ii) multifractal long-term correlated data [MF]. The prescribed fluctuation exponent was $\alpha = 0.9$ in both cases. To each record, we added two types of asymmetric seasonal trends: (i) with only the cycle of seasonal mean and (ii) with the additional cycle

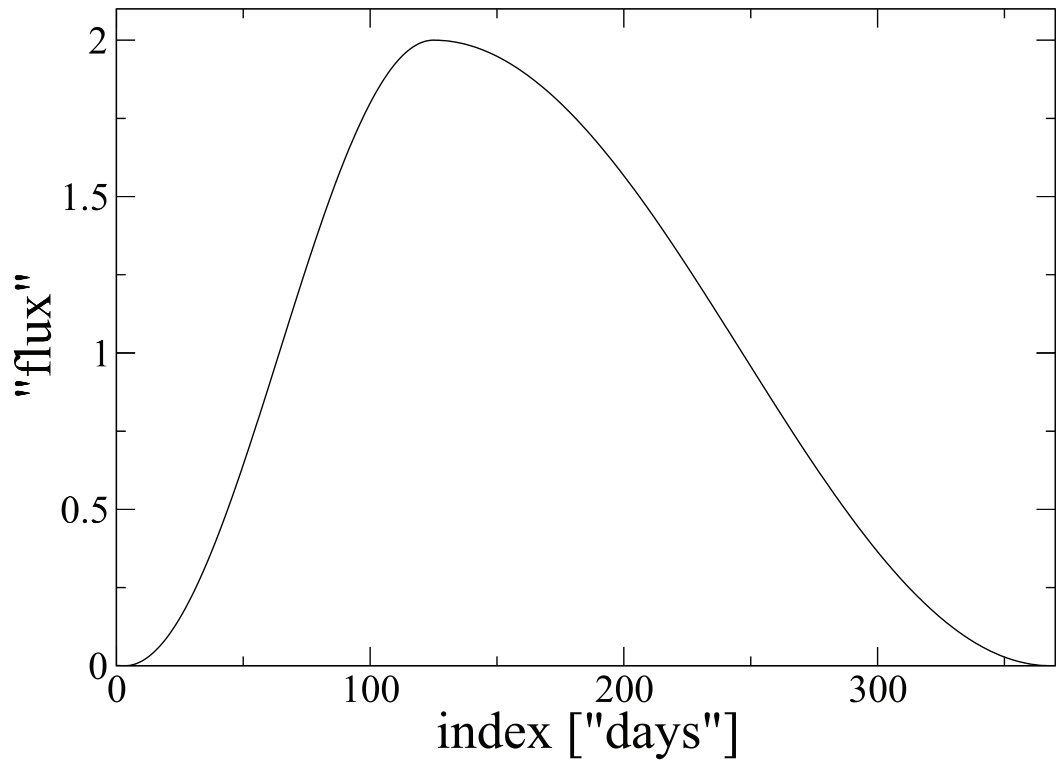


Fig. 13.4. Asymmetric seasonal function C_ℓ, one "year" is shown. The function increases 1/3 of the time period ($T = 365$) and decreases for the rest 2/3 of T.

of standard deviation. The asymmetric annual cycle used for these seasonal trends is defined as

$$C_\ell = C_{k+nT} = \begin{cases} 1 + \cos(2\pi f k), & 0 \leq k < \frac{2}{3}T, \\ 1 - \cos(4\pi f k), & \frac{2}{3}T \leq k < T, \end{cases} \tag{13.10}$$

where n is number of year cycles of period $T = 365\,\text{days}$, $f = 0.75T$, $\ell = 1, \dots\ N-1$. Thus, C_ℓ increases 1/3 of the time period and decreases for 2/3 of T. One seasonal cycle of the function is illustrated in Fig. 13.4.

Using the asymmetric function C_ℓ, we generate four ensembles for testing the filtering techniques as follows:

Data 1. Monofractal long-term correlated process with asymmetric annual cycle of mean

$$D^1_\ell = C_\ell + A_1 \cdot [LTC]_\ell$$

Data 2. Monofractal long-term correlated process with asymmetric annual cycles of mean and standard deviation

$$D^2_\ell = C_\ell + A_2 \cdot C_\ell \cdot [LTC]_\ell + B_2 \cdot [LTC]_\ell$$

Data 3. Multifractal long-term correlated process with asymmetric annual cycle of mean

$$D_\ell^3 = C_\ell + A_3 \cdot [MF]_\ell$$

Data 4. Multifractal long-term correlated process with asymmetric annual cycles of mean and standard deviation

$$D_\ell^4 = C_\ell + A_4 \cdot C_\ell \cdot [MF]_\ell + B_4 \cdot [MF]_\ell,$$

where $A_i,\ B_i$ are rescaling constants.

Each set contained 20 samples of length 65,536 data points. The algorithm for generating monofractal long-term correlated data is described in [13.14], for multifractal long-term correlated data—in [13.19]. The results are shown in Figs. 13.5, 13.6.

In the first column of Fig. 13.5, we present samples of four processes (three-"year" data), DFA curves for raw (green curves), deseasonalised (black curves) and phase-substituted (red curves) data in second column, and DFA curves for volatilities (third column). One can see that DFA curves of deseasonalised and phase-substituted data practically coincide (second column), which means that the seasonal cycle heavily affecting the DFA curve of the raw data (green curves) is successfully removed (in both black and red curves). The DFA curves of volatilities demonstrate a pronounced difference in the cases of Data 2, 4 (where both cycles of seasonal mean and standard deviation are imposed). One can see that the phase-substitution filters out this complicated periodicity better than the conventional deseasonalising.

A difference between two methods can also be observed in Fig. 13.6, where the multifractal exponents and averaged multifractal spectra are compared (deseasonalised data — black curves, phase-substituted data — red curves). By construction, only Data 3, 4 are multifractal, whereas Data 1, 2 are monofractal. If the conventional deseasonalising applied, the artifical multifractality is falsely detected (black spectra), whereas the spectra for phase-substituted data (red curves) are corrupted and narrower in both panels, correctly indicating monofractality. In the case of intrinsically multifractal Data 3, 4, the spectra of the phase-substituted data (red curves) are smooth and wide, which means detection of multifractality — yet they are more narrow than those of deseasonalised data (black curves), where not all seasonal periodicity artefacts are properly removed.

13.5.2 Results of Volatility Analysis of Observed River Data

Having performed the test with artificial data, we apply the developed technique to the observed river flux records. The effect of the proposed phase-substitution filtering technique on the volatility analysis can be demonstrated

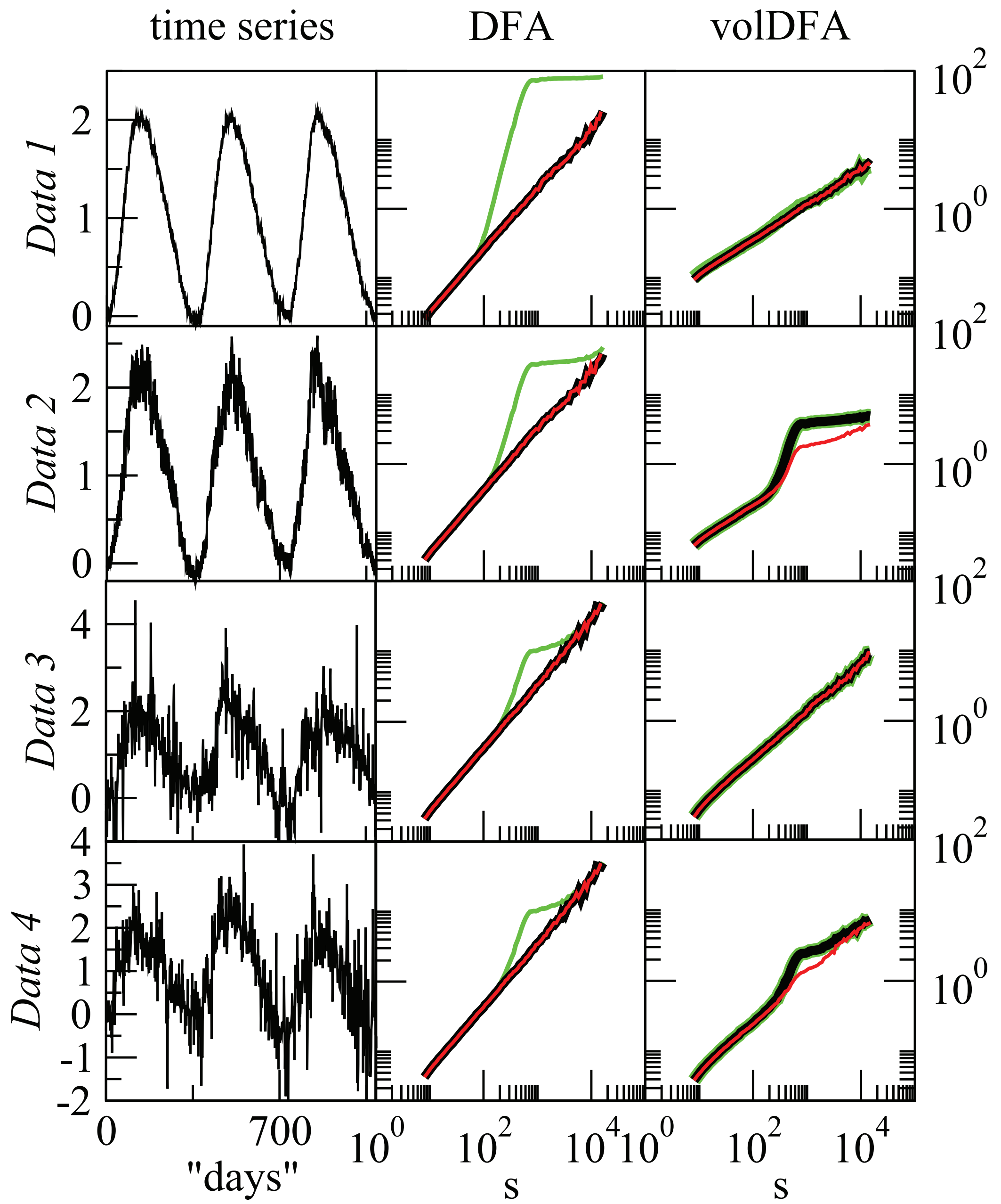


Fig. 13.5. Data and DFA results of the experiment with four ensembles. *Data 1:* monofractal long-term correlated process with asymmetric seasonal cycle of mean values. *Data 2:* monofractal long-term correlated process with asymmetric seasonal cycles of mean and standard deviation. *Data 3:* multifractal long-term correlated process with asymmetric seasonal cycle of mean values. *Data 4:* multifractal long-term correlated process with asymmetric seasonal cycles of mean and standard deviation. Three-year data samples, DFA results, and volatility DFA results (raw data — green curves, deseasonalised data — black curves, phase-substituted data — red curves).

by comparing second moment curves (Fig. 13.2, right panel), where the DFA3

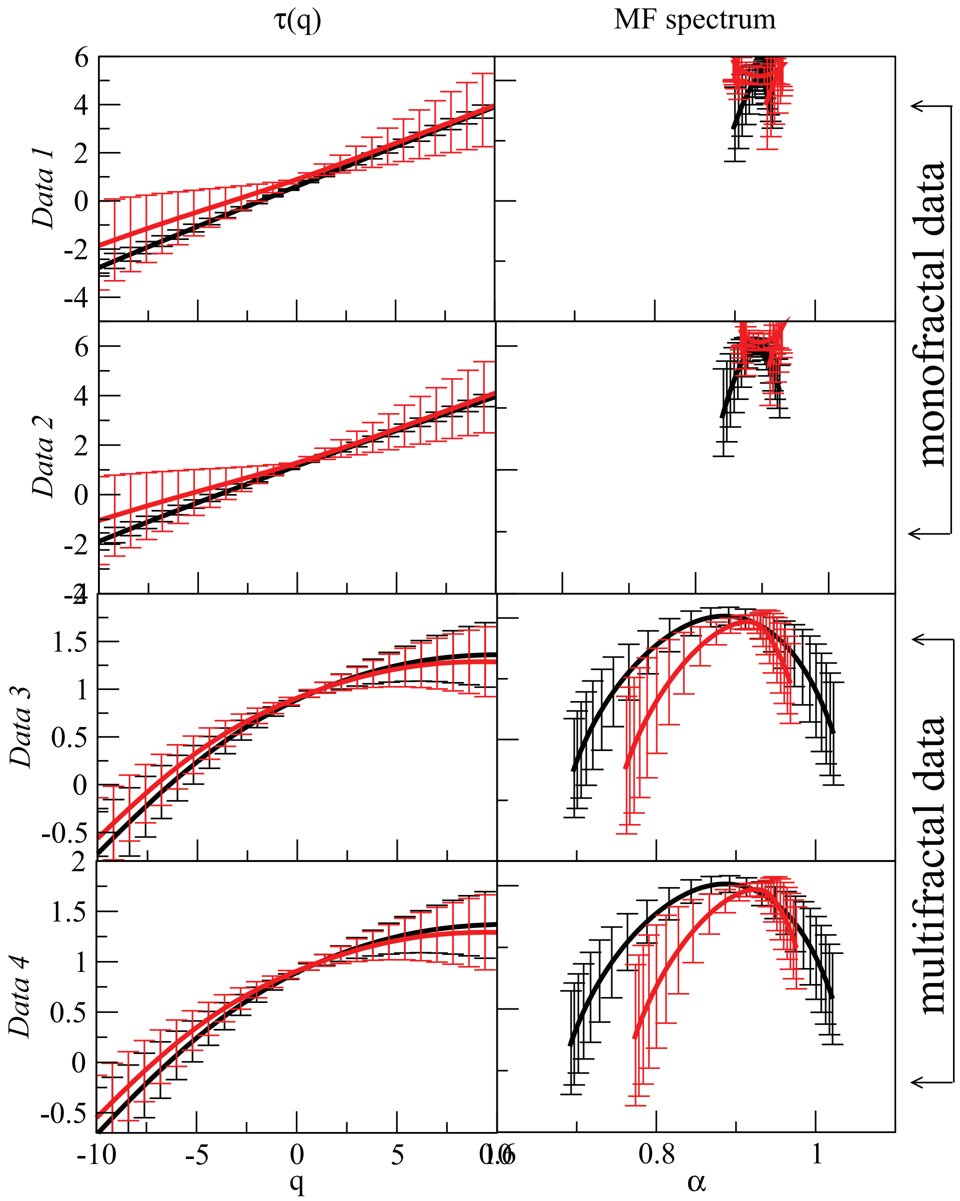


Fig. 13.6. Multifractal results of the experiment with four ensembles of artifical data: exponents $\tau(q)$ and averaged multifractal spectra (deseasonalised data — black curves, phase-substituted data — red curves).

curve of phase-substituted data has no periodicity hump, whereas the curve for the conventionally deseasonalised data does have it.

In Fig. 13.7, we summarize the results of the volatility analysis for thirty world rivers with data preprocessed using the filtering techniques described in Sec. 13.1 (Approaches 1–5). We compare initial time series and their volatilities. For the two-points correlation analysis, the plain raw data show an exponent lower than it should be due to the seasonal periodicity. On the other hand, for the volatility series the filtering techniques that filter out the seasonal

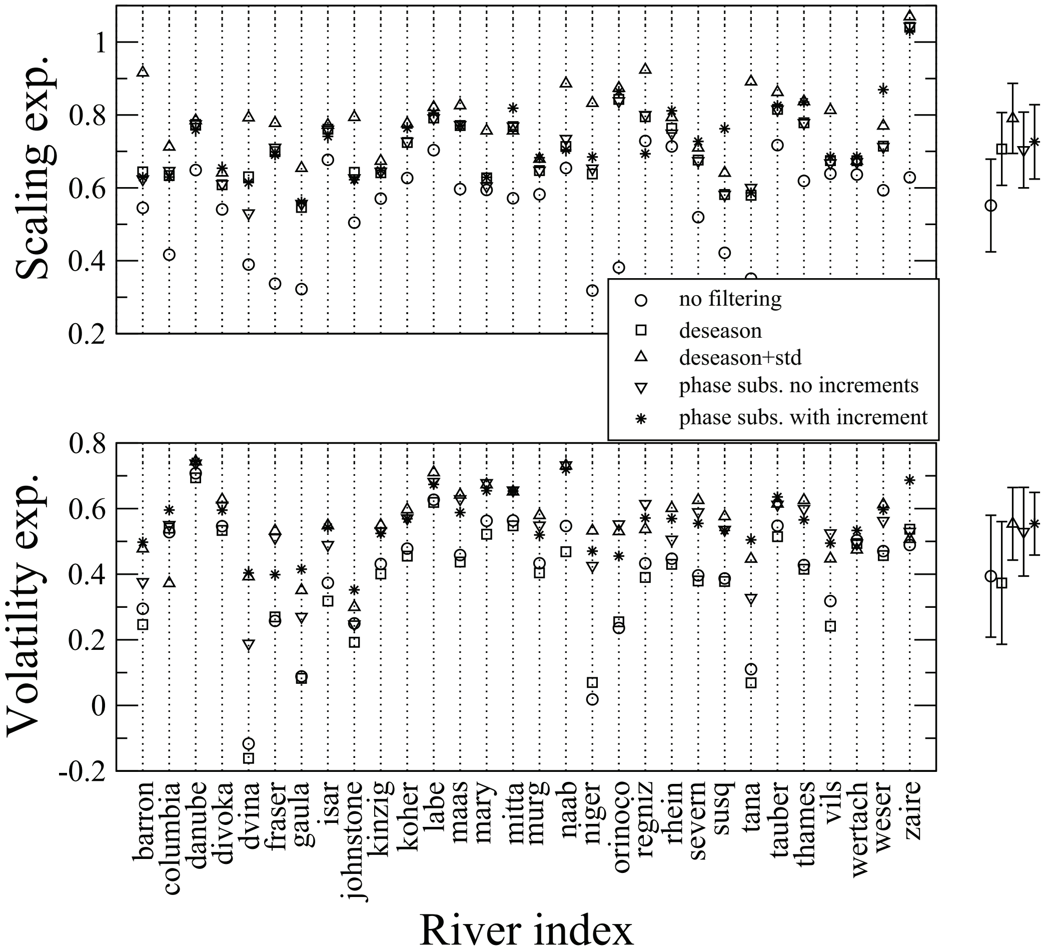


Fig. 13.7. Results of fluctuation analysis for 30 rivers using five filtering approaches. Top panel: exponents in the data, lower panel: exponents in the volatilities; estimated in the asymptotic regime.

standard deviation (Approaches 3–5) show comparable results. This suggests that in order to estimate the long-term nonlinear properties of the hydrological time series, one has to filter out the periodicities associated with the standard deviation.

In our earlier paper [13.22], we obtained an averaged power spectrum exponent of volatilities $\beta \sim 0.27$ for the time scales above one year. Since the DFA fluctuation exponent α is related to the power spectrum exponent β by $\alpha = (1 + \beta)/2$, the power spectrum results correspond to $\alpha \sim 0.63$, which is close to the present results obtained by direct calculation of α in the time scale above one year. Therefore, the earlier suggested method of scaling analysis of volatilities by means of logarithmically binned power spectrum has been proven to be efficient and supports our present results.

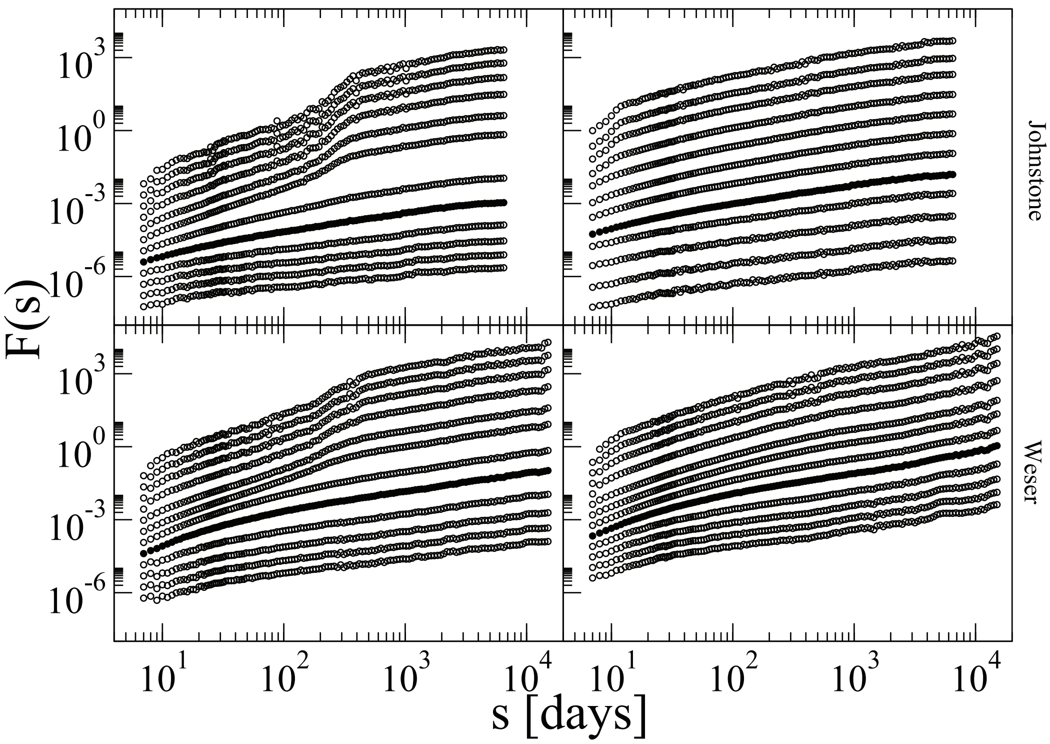


Fig. 13.8. Multifractal partition functions for Johnstone and Weser river flux, q=-10,-8,-6,-4,-2,-1,1,2,4,6,8,10, from top to buttom. Left panels: deseasonalised data; right panels: phase-substituted data. The new filtering technique eliminates humps and crossovers that may lead to erroneous multifractal spectrum.

13.5.3 Results of Multifractal Analysis of Observed River Data

In Fig. 13.8, we show MFDFA partition functions of two river records processed with two filtering techniques. The curves represent discrete values of moments q from -10 to 10, from top to bottom Crossovers in the higher moments curves deviate the results of multifractal analysis (deseasonalised data, left panels). The curves of negative moments are more affected by the periodicities than those of positive moments.

In the right-hand side panels, the curves with filled symbols correspond to DFA3 second moment, and one can see that they are identical to those in the left panels, therefore the filtered data preserve two-point correlations of the deseasonalised data. At the same time, the higher moments' curves have been essentially smoothed (no periodicity humps in curves of negative moments).

Earlier works studied multifractal properties of hydrological time series using the conventional deseasonalising [13.11, 13.25]. To avoid the effect of crossovers in the higher moments' scaling curves (seen in Fig. 13.8, left panel), they calculated the scaling exponents for asymptotic time scales above the "bump" range. In [13.28], the authors applied no preliminary filtering to the data and studied the short-term scaling behaviour. Their results are thus ex-

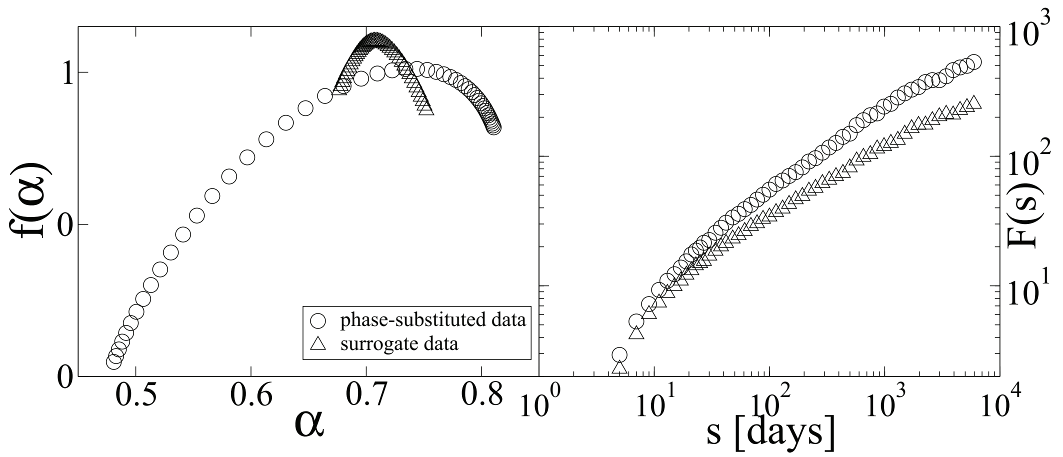


Fig. 13.9. Multifractal spectra (left panel) and DFA3 curves for volatilities (right panel) of phase-substituted (circles) and surrogate (triangles) series. The phase-substitution preserves nonlinear properties of the data which can be eliminated by randomizing the phases.

pected to be influenced by the periodicity artefacts, although their general conclusion that river flux fluctuations are multifractal is most probably valid. The method proposed here may enable extending the region of scaling that includes the annual time scale (the "bump" region), because the crossovers discussed above do not exist after applying our filtering procedure.

In Fig. 13.9, we compare multifractal spectra (left panel) and volatilities (right panel) of phase-substituted and phase-randomized data for the Johnstone river. One can see that the phase-substituted data is nonlinear and that the nonlinearities are diminished by the surrogate test. This effect supports our filtering technique, because it indicates that the phase substitution preserves the essential nonlinear features of the record.

In Fig. 13.10, we compare multifractal spectra of conventionally deseasonalised and phase-substituted data of six rivers. Results for the phase-substituted data demonstrate weakened nonlinearity: the spectra become narrower.

Having performed the multifractal analysis of the phase-substituted data at time scales above one year, we have obtained the multifractal spectra of width 0.34 ± 0.26. For the conventionally deseasonalised data, in the time scale above one year we obtained spectra of width 0.49 ± 0.28. Since the error-bars are quite wide, we can conclude that both methods provide close results in the asymptotic scale. We hypothesize that if one would consider a scaling range that includes the crossover, the difference between the two filtering techniques would have become more significant. For that, we measure spectra width in a longer time scale range, above 100 days, where the scaling curves for conventionally deseasonalised data have the periodicity crossovers. For the phase substituted data, in time scales above 100 days the spectra width for the thirty rivers is 0.46 ± 0.13, as compared to the results using the conventionally

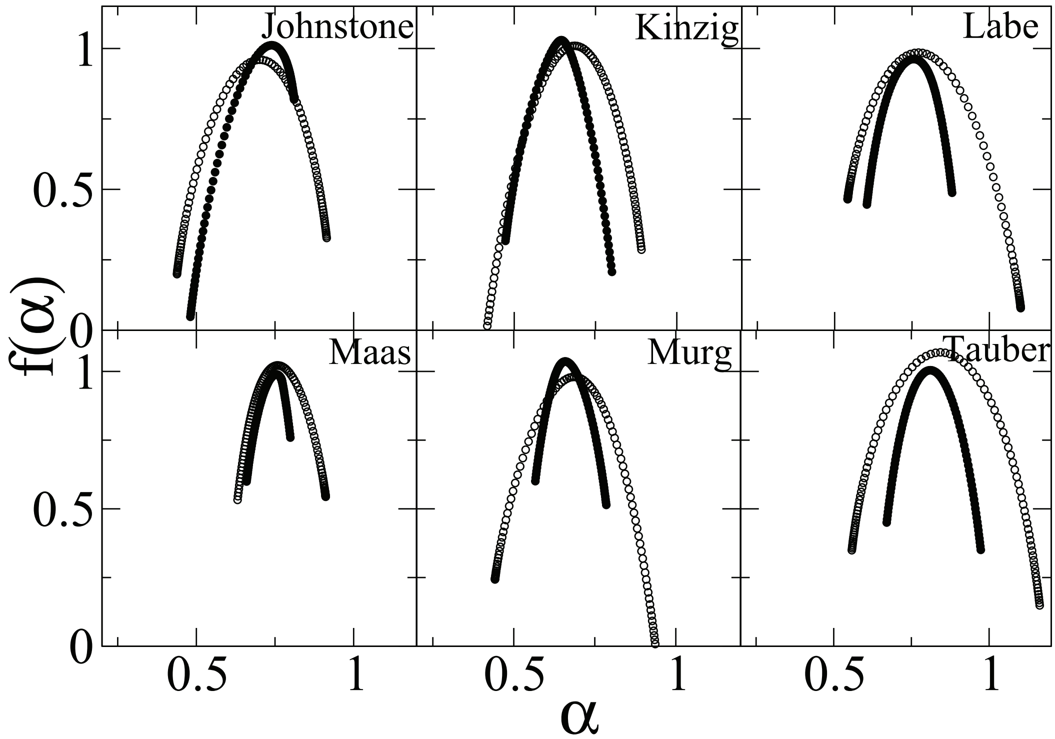


Fig. 13.10. Multifractal spectra of six rivers, compared deseasonalised (open symbols) and phase-substituted (filled symbols) data (the Johnstone river - Australia, 74 yr; the Kinzig river - Germany, 82 yr; the Labe river - Czechia, 102 yr; the Maas river - Netherlands, 80 yr; the Murg river - Germany, 77 yr; the Tauber river - Germany, 66 yr). The spectra of phase-substituted data are narrower: less influenced by the periodicity artefacts in the data.

deseasonalised approach, 0.68 ± 0.20. We note however, that even though the periodicities where filtered out, the wider MF spectrum we obtain for windows scales larger than 100 days may be attributed to the crossovers in the scaling curves.

13.6 Conclusion

Periodicities in climate records might influence the estimation of the two–point scaling exponent, as well as nonlinear properties of hydrological time series, such as volatility correlations and multifractal spectrum width. We compared: (i) raw data, (ii) data where the seasonal average was subtracted, (iii) data where seasonal average was subtracted and the series was then divided by the seasonal standard deviation, (iv) phase-substituted data without taking increment, and (v) phase-substituted data after taking the increments. We found that in order to accurately estimate two-point correlations, it is necessary to apply at least approach (ii), while in order to accurately estimate the long-range nonlinear properties (the multifractal spectrum and volatility correlations), approaches (iii)-(iv) may be applied. These approaches basically filter out the periodicity in the standard deviation.

The proposed filtering technique is merely empirical and was developed for studying hydrological time series with specific scaling properties. However, the technique may also be useful for estimation of nonlinear properties in other climatic time series with similar statistical characteristics.

Acknowledgments. The authors thank the German Federal Ministry of Science for financial support (grant no. 03F0205B9 and 01LD0016). We would like to thank J. Kantelhardt for useful discussions. YA is supported by the Yeshaya Horowitz Association in the Center for Complexity Science.

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